Meeting 1119

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Research motivation

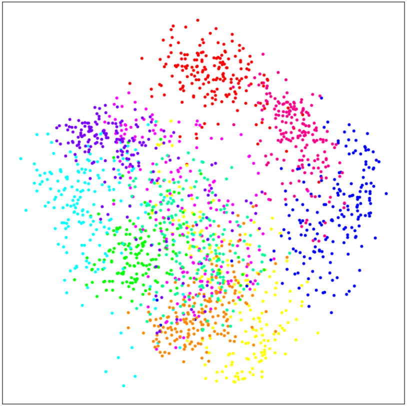
Semiconductor Manufacturing Process

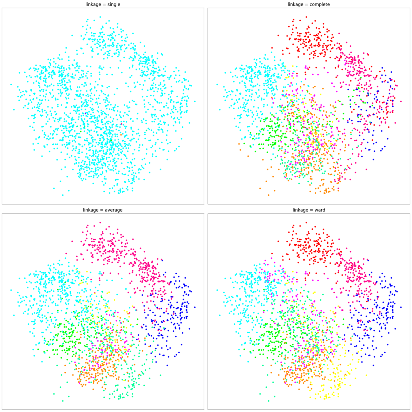
- Unbox $y = \hat{f}(x)$
- Solution to imbalanced data (differ from SMOTE)
- Fitting model

Research method

- Features selection
  - PCA
  - NR(p-value)
  - RF(importance)
- Clustering
  - KMeans
  - AgglomerativeClustering(Hierarchical clustering)
- Model training(StandardScaler, SMOTE)
  1. LR, SVM, DT, RF, XGB
  2. ACC, FAR

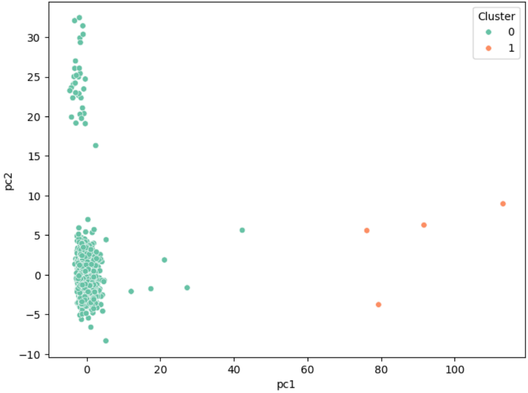
Research method AgglomerativeClustering





Results I PCA + KMeans, 2 clusters

cluster	0	1
$y = 0$	1460	3
$y = 1$	103	1



Results I　Training in cluster 0

LR

$$\begin{bmatrix} 265 & 27 \\ 15 & 6 \end{bmatrix}$$

SVM

$$\begin{bmatrix} 292 & 0 \\ 21 & 0 \end{bmatrix}$$

DT

$$\begin{bmatrix} 259 & 33 \\ 13 & 8 \end{bmatrix}$$

RF

$$\begin{bmatrix} 292 & 0 \\ 20 & 1 \end{bmatrix}$$

XGB

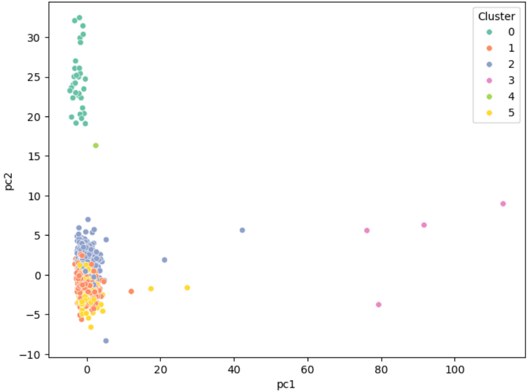
$$\begin{bmatrix} 291 & 1 \\ 19 & 2 \end{bmatrix}$$

Summary

model	ACC	FAR
LR	0.8658	0.0925
SVM	0.9329	0.0000
DT	0.8530	0.1130
RF	0.9361	0.0000
XGB	0.9361	0.0034

Results II PCA + KMeans, 6 clusters

cluster	0	1	2	3	4	5
$y = 0$	25	563	627	3	1	244
$y = 1$	7	36	52	1	0	8



Results II Training in cluster 2

LR SVM DT

$$\begin{bmatrix} 110 & 16 \\ 7 & 3 \end{bmatrix} \quad \begin{bmatrix} 126 & 0 \\ 10 & 0 \end{bmatrix} \quad \begin{bmatrix} 106 & 20 \\ 5 & 5 \end{bmatrix}$$

RF XGB

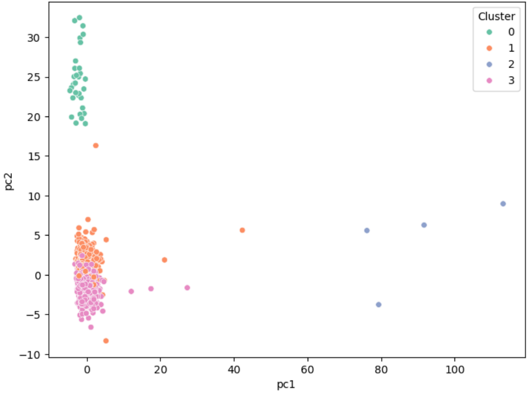
$$\begin{bmatrix} 126 & 0 \\ 9 & 1 \end{bmatrix} \quad \begin{bmatrix} 122 & 4 \\ 8 & 2 \end{bmatrix}$$

Summary

model	ACC	FAR
LR	0.8309	0.1270
SVM	0.9265	0.0000
DT	0.8162	0.1587
RF	0.9338	0.0000
XGB	0.9118	0.0317

Results III Standard + KMeans, 4 clusters

cluster	0	1	2	3
$y = 0$	25	667	3	768
$y = 1$	7	55	1	41



Results III Training in cluster 1

LR

$$\begin{bmatrix} 117 & 17 \\ 9 & 2 \end{bmatrix}$$

SVM

$$\begin{bmatrix} 134 & 0 \\ 11 & 0 \end{bmatrix}$$

DT

$$\begin{bmatrix} 116 & 18 \\ 8 & 3 \end{bmatrix}$$

RF

$$\begin{bmatrix} 134 & 0 \\ 11 & 0 \end{bmatrix}$$

XGB

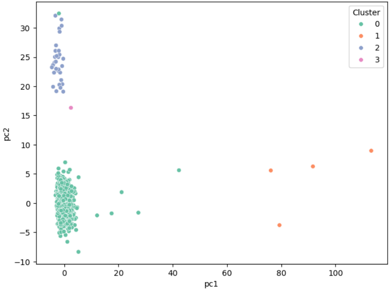
$$\begin{bmatrix} 130 & 4 \\ 10 & 1 \end{bmatrix}$$

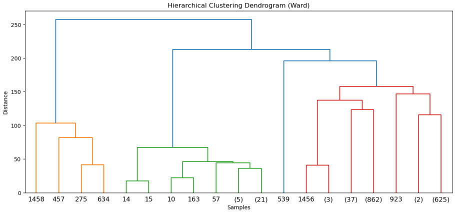
Summary

model	ACC	FAR
LR	0.8207	0.1269
SVM	0.9241	0.0000
DT	0.8207	0.1343
RF	0.9241	0.0000
XGB	0.9034	0.0299

Results IV PCA + AgglomerativeC, 4 clusters

cluster	0	1	2	3
$y = 0$	1435	3	24	1
$y = 1$	96	1	7	0





Results IV Training in cluster 0

LR

$$\begin{bmatrix} 256 & 32 \\ 14 & 5 \end{bmatrix}$$

SVM

$$\begin{bmatrix} 288 & 0 \\ 19 & 0 \end{bmatrix}$$

DT

$$\begin{bmatrix} 266 & 22 \\ 18 & 1 \end{bmatrix}$$

RF

$$\begin{bmatrix} 288 & 0 \\ 19 & 0 \end{bmatrix}$$

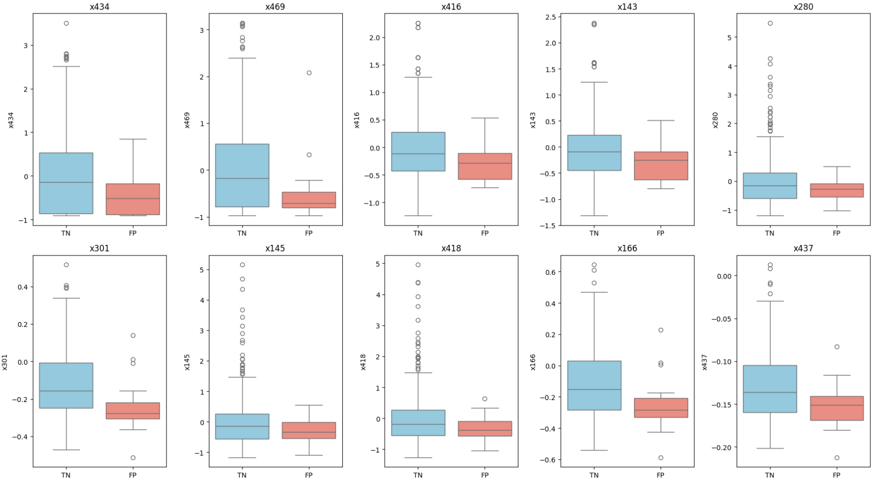
XGB

$$\begin{bmatrix} 288 & 0 \\ 19 & 0 \end{bmatrix}$$

Summary

model	ACC	FAR
LR	0.8502	0.1111
SVM	0.9381	0.0000
DT	0.8697	0.0764
RF	0.9381	0.0000
XGB	0.9381	0.0000

Results IV Some plots



References

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The End

Questions & Comments